

# MQTT Subsystem P

|                    |                |
|--------------------|----------------|
| Team Number:       | 305            |
| Project Name:      | MQTT Subsystem |
| Team Member Names: | Damian         |
| Version:           |                |

## A. List ALL major components (active devices, integrated circuits, etc.) except for power source

| All Major Components | Component Name | Part Number   | yVoltageRan | # |
|----------------------|----------------|---------------|-------------|---|
|                      | ESP32          | ESP32-S3-WROC | 2.3V-3.6V   | 1 |

## B. Assign each major component above to ONE power rail below. Try to minimize the number of power rails

| +3.3V Power Rail | Component Name | Part Number   | yVoltageRan | # |
|------------------|----------------|---------------|-------------|---|
|                  | ESP32          | ESP32-S3-WROC | 2.3V-3.6V   | 1 |
|                  | Red LED        | 150060SS75000 | 1.9V-2.4V   | 1 |
|                  | Yellow LED     | 150060YS75000 | 2V-2.4V     | 1 |

**c4. Regulator or Source Ch** +3.3V low-dropout regula LM2575-3.3WU- +4V - 40V 1  
**Total Rem**

## C. For each power rail above, select a specific voltage regulator using the same process

**+3.3V Power Rail -> LM2575-3.3**

## D. Select a specific external power source (wall supply or battery) for your system, and

| External Power Source 1  | Component Name      | Part Number      | yVoltageRan                      | Output |
|--------------------------|---------------------|------------------|----------------------------------|--------|
| Power Source 1 Selection | Plug-in Wall Supply | full part number | 110VAC                           | +24V   |
|                          | +3.3V Regulator     | LM2575-3.3WU-1   | +4V-40V                          | +3.3V  |
|                          |                     |                  | <b>Total Remaining Current A</b> |        |

## Notes

External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator. If you have multiple units in your design (e.g., a base unit and remote unit) then you need a supply for each unit.

# Power Budget

| Power sources, voltage regulators, resistors, capacitors, or passive elements |                  |      |
|-------------------------------------------------------------------------------|------------------|------|
| AbsoluteMaximumCurrent (mA)                                                   | TotalCurrent(mA) | Unit |
| 1000                                                                          | 1000             | mA   |

| Number of different power rails in the design.  |                  |      |
|-------------------------------------------------|------------------|------|
| AbsoluteMaximumCurrent (mA)                     | TotalCurrent(mA) | Unit |
| 500                                             | 500              | mA   |
| 20                                              | 20               | mA   |
| 20                                              | 20               | mA   |
|                                                 | 0                | mA   |
| <i>Subtotal</i>                                 | 540              | mA   |
| <i>Safety Margin</i>                            | 25%              |      |
| <i>Total Current Required on +3.3V Rail</i>     | 675              | mA   |
| 1000                                            | 1000             | mA   |
| <i>Remaining Current Available on 3.3V Rail</i> | 325              | mA   |

Assess as for major component selection. Confirm that the Total Remaining

and confirm that it can supply all of the regulators for all of the power rails

| AbsoluteMaximumCurrent (mA)                 | TotalCurrent(mA) | Unit |
|---------------------------------------------|------------------|------|
| 5000                                        | 5000             | mA   |
| 1000                                        |                  |      |
| <i>Available on External Power Source 1</i> | 4000             | mA   |

regulator (e.g., +14V for a +12V regulator).  
 Prepare power budget for each unit